IS Course 4 Database Modeling

1- Database Modeling and Design Steps

- *Data modeling* is the *first step* of the database design process. High-level (entity-relationship)
  - This step is sometimes considered a high-level *abstract design* phase, also called *conceptual design*.
  - The objective of this phase is to describe:
    - The data contained in the database (e.g., entities: students, professors, courses, subjects)
    - The relationships between data elements (e.g., students are supervised by professors; professors teach courses)
    - The constraints on the data (e.g., the student ID has exactly eight digits; a subject has only four or six credits)
- *The second step* is to express the data elements, relationships, and constraints using the concepts provided by the high-level data model.
  - Because these concepts do not include implementation details, the result of the data modeling process is a (semi-)formal representation of the database structure.
  - This result is fairly easy to understand, so it is used as a reference to ensure that all user requirements are met.
- *The third step* is database design.
  - During this step, we can have two sub-steps:
    - One called logical database design, which defines a database within a specific DBMS data model (internal, all three types, including relational)
    - and another called physical database design, which defines the database's internal storage structure, file organization, or indexing techniques.
  - These two sub-steps are the steps for implementing the database and creating user operations/interfaces.

2- Defining a Data Model

- In the database design phase, data is represented using a specific data model.
- A data model is a collection of concepts or notations for describing data, relationships between data, data semantics, and data constraints.
- Most data models also include a set of basic operations for manipulating data in the database.

3- Degrees of Data Abstraction

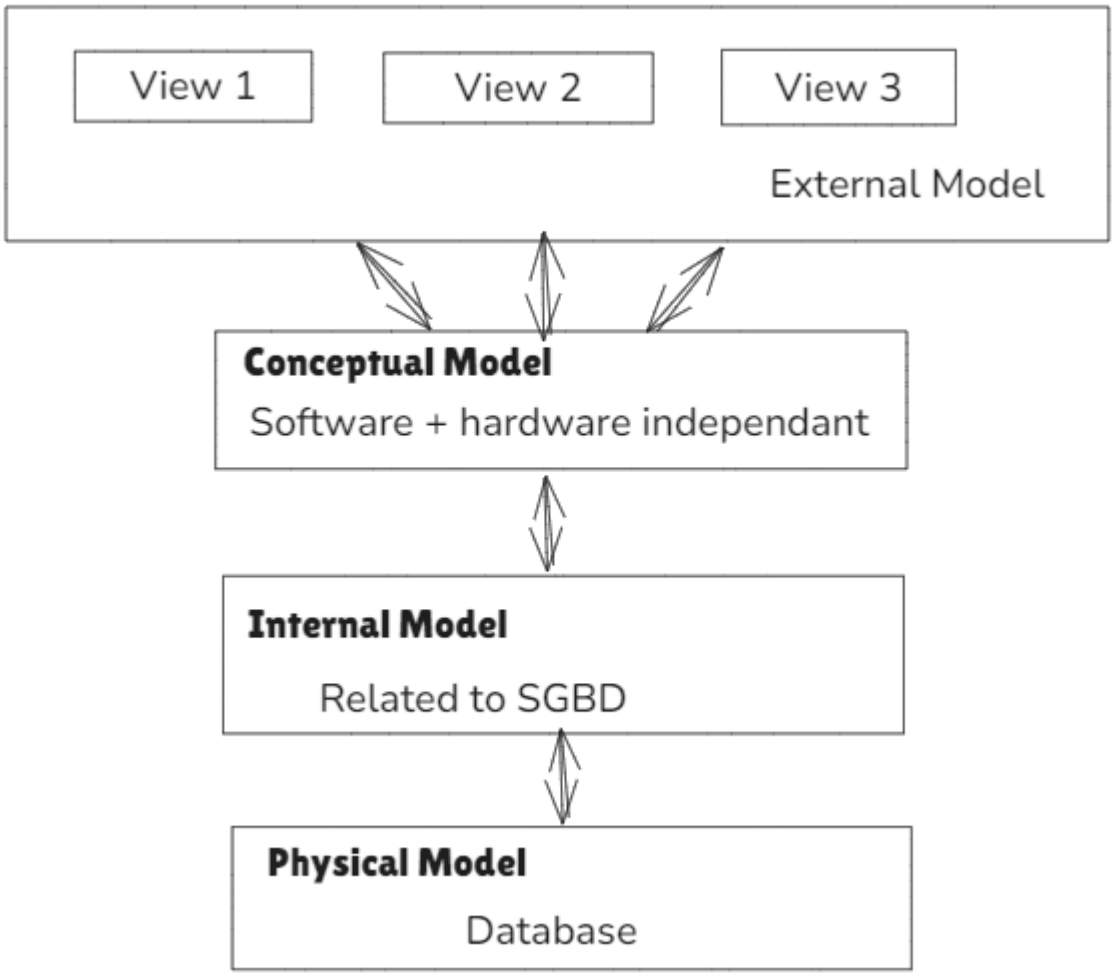
There are four types of data models: external, conceptual, internal, and physical.

- External models represent the user's view of the database.
- Conceptual models provide flexible data structuring capabilities.

- Internal models view a database as a collection of fixed-size records (relational data model, network data model, hierarchical data model).
- Physical models: Represent the physical representation of the database. They have the lowest level of abstraction and describe how data is stored.

4- Data Abstraction Layer

- The data abstraction layer hides the complexity of data from database users.
- It provides a consistent interface to the data, regardless of how the data is physically stored and organized.
- The data abstraction layer allows application developers to focus on the business logic of their applications, rather than the details of data storage.



5- Types of Record-Based Logical Data Models

- These models provide concepts that users can understand, but they are not too far removed from how data is stored on a computer.
- Three well-known data models of this type are relational data models, network data models, and hierarchical data models.

5.1 Relational Model

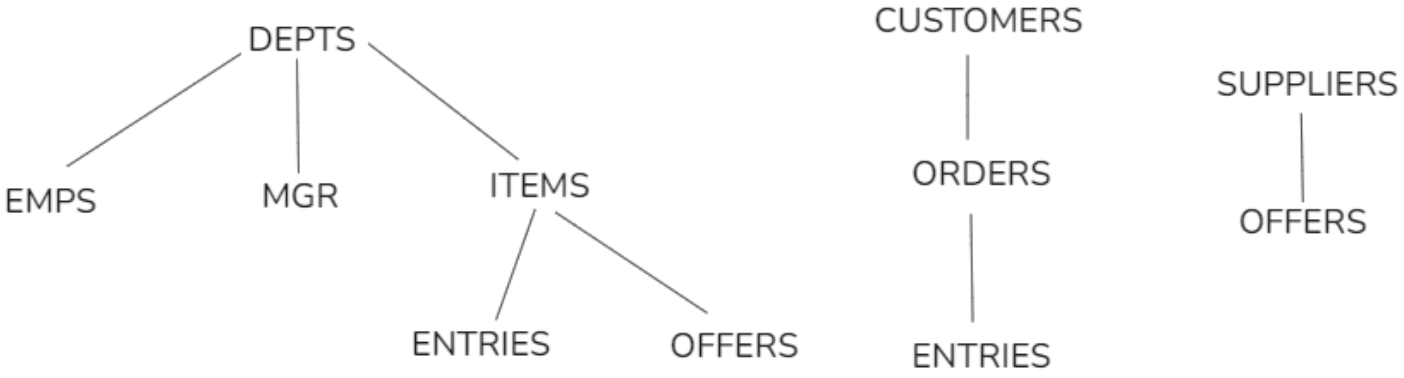
- This model represents data as relationships or tables.
- For example, in the Science World membership system, each membership has many members.

- The membership ID, expiration date, and address information are fields in the membership.
- Members are individuals such as Mickey, Minnie, Mighty, Door, Tom, King, Man, and Moose.
- Each record is said to be an instance of the membership table. - terms used: relation or table, fields-attributes, record: record (row, tuple) instance

customer_id	store_id	first_name	last_name	email
1	1	MARY	SMITH	MARY.SMITH@sakilacustomer.org
2	1	PATRICIA	JOHNSON	PATRICIA.JOHNSON@sakilacustomer.org
3	1	LINDA	WILLIAMS	LINDA.WILLIAMS@sakilacustomer.org
4	2	BARBARA	JONES	BARBARA.JONES@sakilacustomer.org
5	1	ELIZABETH	BROWN	ELIZABETH.BROWN@sakilacustomer.org
6	2	JENNIFER	DAVIS	JENNIFER.DAVIS@sakilacustomer.org
7	1	MARIA	MILLER	MARIA.MILLER@sakilacustomer.org

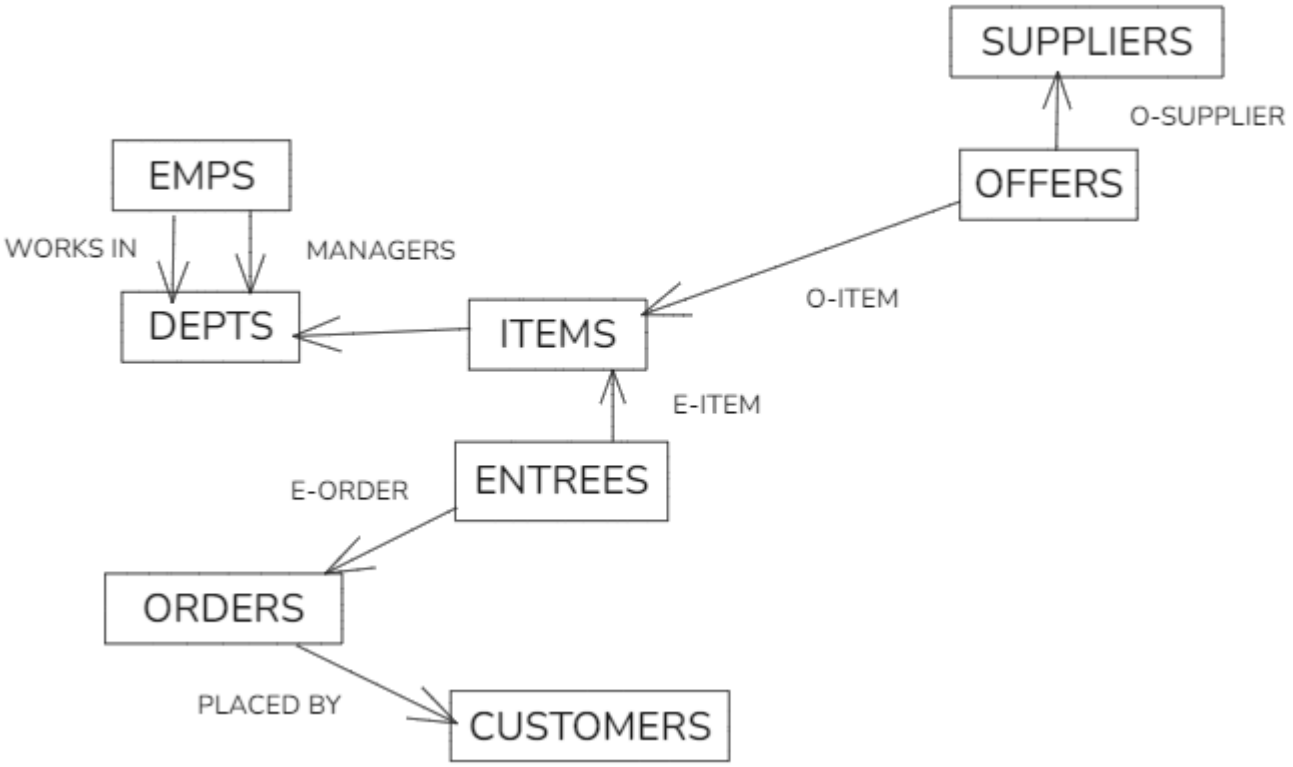
5.2 Hierarchical Model

- This model represents data in a hierarchical tree structure.
- Each branch of the hierarchy represents a number of related records.



5.3 Network Model

- This model represents data as record types.
- This model also represents a limited type of one-to-many relationship called a set type
- Terms used: record, set



6- The Relational Data Model

The relational data model was introduced by E. F. Codd in 1970. It is currently the most widely used data model. The relational model has provided the basis for:

- Research on data/relationship/constraint theory
- Many database design methodologies
- The standard database access language called Structured Query Language (SQL)
- Nearly all modern commercial database management systems

The relational data model describes the world as "a set of interdependent relationships (or tables)."

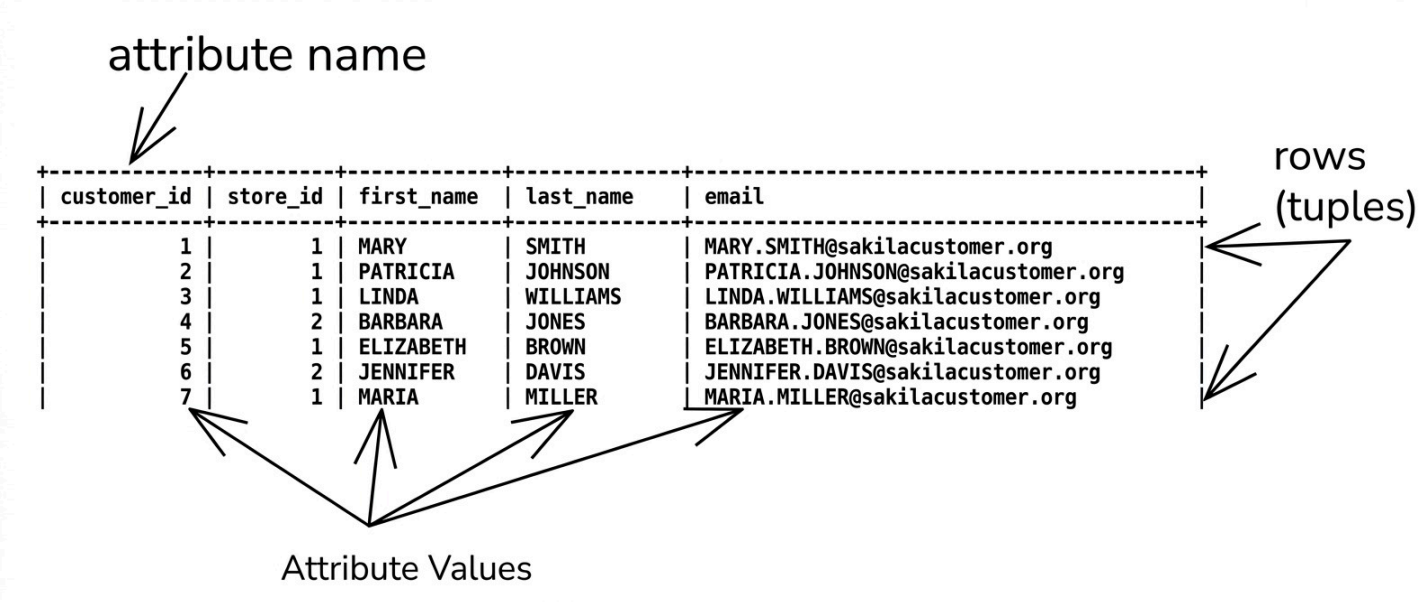
6-1 Fundamental Concepts of the Relational Data Model

- **Relation = Table**
  - A relation, also called a table or file, is a subset of the Cartesian product of a named list of domains.
  - In a table, each row represents a group of related data values. - A row, or record, is also called a tuple.
  - A column in a table is a field and is also called an attribute.
  - *Formal definition:*

$$\begin{aligned}&\text{- Given } n \text{ domains}\\&\text{- And } r \text{ a relation defined on these domains}\\&\text{- Then: } r \subseteq D_1 \times D_2 \times \ldots D_n\end{aligned}$$

- **Table**:
  - A database is composed of multiple tables, and each table holds the data
- **Column**:

    - The main storage unit in a database is called a column, field, or attribute.
    - They house the basic data components into which your content can be decomposed.
    - When deciding which fields to create, you can, for example:
        - Select the common components of the information you will store in the database
        - Avoid details that distinguish one element from another.
- **Domain**
    - A domain is the original set of atomic values used to model data.
    - An atomic value means that each value in the domain is indivisible with respect to the relational model.
    - Examples:
        - The First Name domain is the set of strings that represent people's names.
        - The Marital Status domain offers a set of possibilities: Married, Single, Divorced.
- **Records**
    - Records contain fields that are related, such as a customer or an employee.
    - As mentioned earlier, a tuple is another term used for record. (or a line)



- **Degree**
    - The degree is the number of attributes in a table. In the example above, the degree is 5.

6-2 Table Properties

- A table has a name that is distinct from all other tables in the database.
- There are no duplicate rows; each row is distinct.
- Column entries are atomic. The table does not contain repeating groups or multi-valued attributes.
- Column entries belong to the same domain based on their data type, including:
- number (numeric, integer, float, smallint, etc.)
- character (string)
- date
- logical (true or false)

- Operations combining different data types are not allowed.
- Each attribute has a distinct name.
- The sequence of columns is not significant.
- The sequence of rows is not significant.

Bibliographic References

Watt, Adrienne, and Nelson Eng. *Database design*. 2nd Edition. BCcampus, 2014